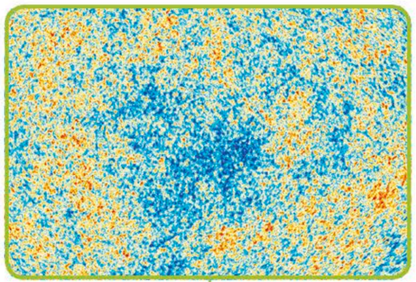
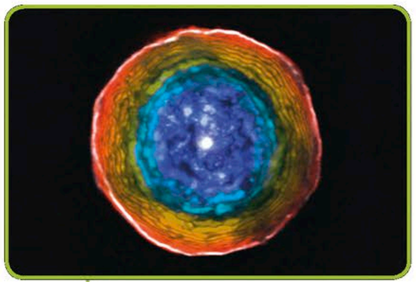
All electromagnetic waves travel through space at the **speed of light**, which is **299,792,458 m/s** (commonly rounded to 300,000 km/s).

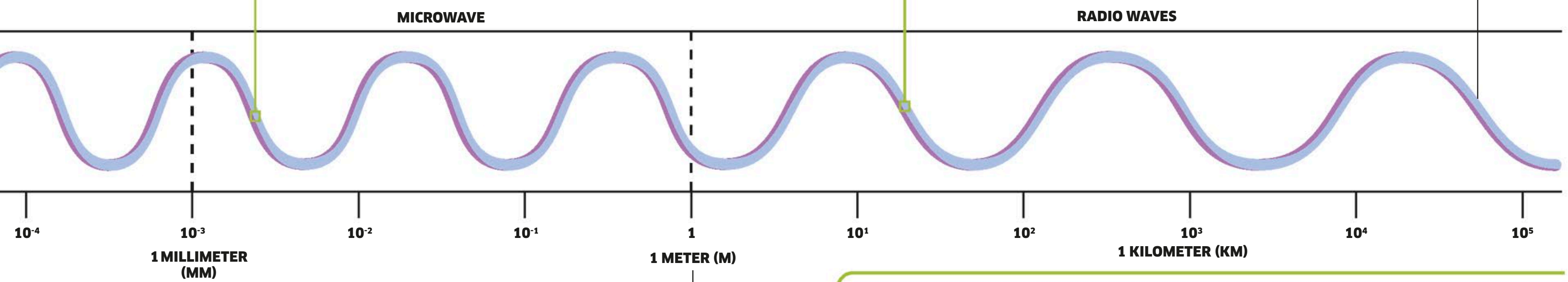
Extremely low frequency (ELF) radio waves are used to **communicate with submarines.**



Microwaves
On Earth, microwaves are used for radar, cell phone, and satellite communications. Scientists have captured images (left) of microwaves left over from the Big Bang at the birth of the universe.



Radio waves
The longest waves on the spectrum, radio waves carry TV as well as radio signals. Radio telescopes are able to capture radio waves emitted by sources in space and convert them into images, such as this star (left).

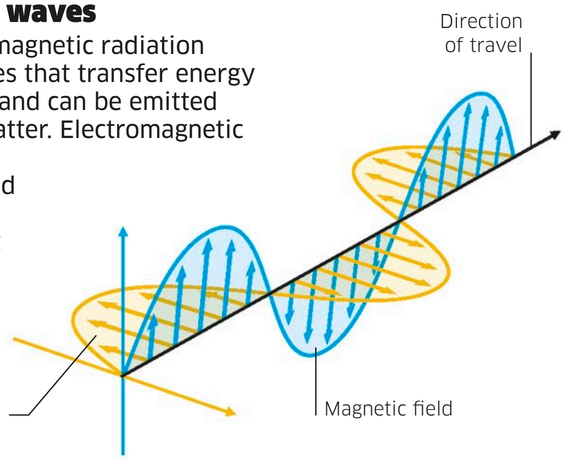


The electromagnetic spectrum

There are electromagnetic waves over a wide range of wavelengths, from gamma waves, which have the shortest wavelength and highest energy, to radio waves, which have the longest wavelength and lowest energy. All electromagnetic waves are invisible, except for those that make up light. As you move along the spectrum, different wavelengths are used for a variety of tasks, from sterilizing food and medical equipment to communications.

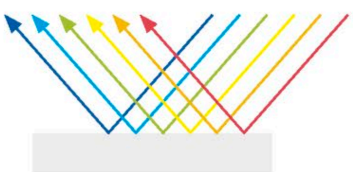
Electromagnetic waves

All types of electromagnetic radiation are transverse waves that transfer energy from place to place and can be emitted and absorbed by matter. Electromagnetic radiation travels as waves of electric and magnetic fields that oscillate (vibrate) at right angles to each other and to the direction of travel.



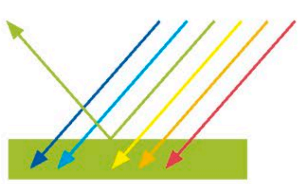
Seeing color

We see color based on information sent to the brain from light-sensitive cells in the eye called cones. There are three types of cone, which respond to red, green, or blue light. We see all colors as a mix of these three colors. Objects reflect or absorb the different colors in white light. We see the reflected colors.



White object
White objects reflect all the colors that make up the visible light spectrum, which is why they appear white.

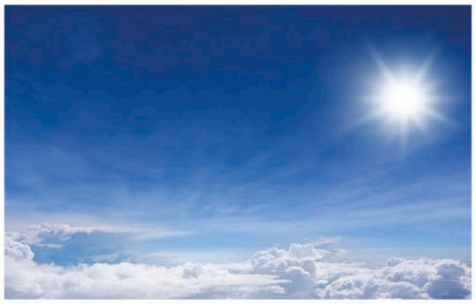
Black object
Black objects absorb all the colors of the visible light spectrum and reflect none. They also absorb more heat.



Green object
We see green objects because they reflect only the green wavelengths of visible light.

Light scattering

When sunlight hits Earth's atmosphere, air molecules, water droplets, and dust particles scatter the light, but they don't scatter the colors equally. This is why the sky is blue, clouds are white, and sunsets red.



Blue sky
The blue of the sky is caused by air molecules in the atmosphere, which scatter short-wavelength light at the blue end of the spectrum. Larger water and dust particles scatter the full spectrum as white light. The bluer the sky, the purer the air.

Red sunset
When the sun is low in the sky, light takes a longer path through the atmosphere, more light is scattered, and shorter wavelengths are absorbed. At sunrise and sunset, clouds may appear red, reflecting the color of light shining on them.